

CYBERSECURITY MASTER REPORT

Summit Care Medical Clinic Cyber Threat Analysis & Risk Assessment

A complete rebranded security portfolio document combining the Summit Care healthcare network blueprint with the prior cyber threat analysis project. This report presents a realistic, sanitized assessment of penetration testing, vulnerability management, HIPAA-aware controls, third-party risk, audit evidence, incident response, and remediation planning for a mock medical clinic network.

Penetration Testing

Risk Assessments

HIPAA Security Rule Alignment

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CYBERSECURITY FRAMEWORK

10.7.0.0/16

Sanitized Summit Care private network model

6

NIST CSF 2.0 functions used for mapping

12

Representative findings and audit scenarios

90 days

Priority remediation roadmap

DOCUMENT INDEX

Table Of Contents

1. Executive Summary**2. Engagement Profile****3. Summit Care Environment****4. Methodology****5. Asset & Data Classification****6. Threat Model****7. Penetration Testing Narrative****8. Risk Register****9. HIPAA & Compliance Alignment****10. Third-Party Audit Program****11. Monitoring & Routine Auditing****12. Incident Response & Recovery****13. Remediation Roadmap****14. Evidence & References**

EXECUTIVE SUMMARY

Security Assessment Summary For Reviewers

This master report reframes the existing Summit Care Medical Clinic network architecture into a cybersecurity assessment deliverable. The goal is to show the full security workflow: understanding the business environment, identifying critical healthcare assets, testing realistic attack paths, ranking risk, recommending controls, documenting HIPAA-aware safeguards, and preparing the organization for continuous improvement.

Summit Care is a fictionalized healthcare clinic used to protect real client identities while preserving the technical substance of infrastructure and security work. The environment includes segmented clinical, administrative, billing, voice, wireless, and guest networks; Microsoft identity services; endpoint standards; network monitoring; backup planning; and hybrid cloud collaboration through Microsoft 365 and Entra-connected services.

High

Top residual risk: exposed management and weak administrative access paths

Critical

Protected health information and billing systems are crown-jewel assets

Moderate

Overall maturity after segmentation, MFA, backups, and monitoring baseline

Continuous

Quarterly testing and recurring audit cadence recommended

Portfolio positioning: This report demonstrates cybersecurity analysis, risk assessment, audit documentation, penetration testing workflow, technical writing, healthcare compliance

awareness, and security governance in one organized master document.

ENGAGEMENT PROFILE

Scope, Assumptions, And Rules Of Engagement

The assessment was modeled as an authorized internal security review for a small healthcare office preparing for stronger network governance and HIPAA Security Rule alignment. Testing scenarios are written as controlled simulations and sanitized examples, not instructions for unauthorized activity.

In Scope

- Clinical systems, EMR-facing servers, billing applications, admin workstations, and shared services.
- Firewall rules, VLAN segmentation, wireless access, VPN access, DNS, DHCP, AD, NPS/RADIUS, and endpoint posture.
- HIPAA Security Rule safeguards, NIST CSF 2.0 mapping, incident response, backup readiness, and third-party risk.

Out Of Scope

- Unsafe production exploitation, destructive testing, denial-of-service activity, and live PHI exposure.
- Unapproved password harvesting, persistence deployment, or changes to patient-care systems during business hours.
- Real patient names, production credentials, public IP addresses, vendor secrets, or client-identifying data.

Engagement Area	Purpose	Representative Output
Risk Assessment	Identify threats, vulnerabilities, likelihood, impact, and control gaps affecting ePHI and operations.	Risk register, heatmap, remediation priority, executive summary.

Engagement Area	Purpose	Representative Output
Penetration Testing	Safely validate whether identified weaknesses can become practical attack paths.	Attack narrative, proof summaries, screenshots or logs in sanitized form, retest results.
Compliance Review	Map controls to HIPAA administrative, physical, and technical safeguards.	Safeguard matrix, policy gaps, audit evidence checklist.
Third-Party Audit	Assess vendors supporting EMR, billing, cloud storage, VoIP, remote support, and managed services.	Vendor risk scores, contract/security questionnaire notes, corrective action tracking.

SUMMIT CARE NETWORK

Healthcare Network Security Context

Summit Care's mock environment represents a small but realistic medical clinic: clinicians need fast access to patient records, billing staff handle financial workflows, administrators manage scheduling and documents, and IT must protect the network without slowing down patient care.

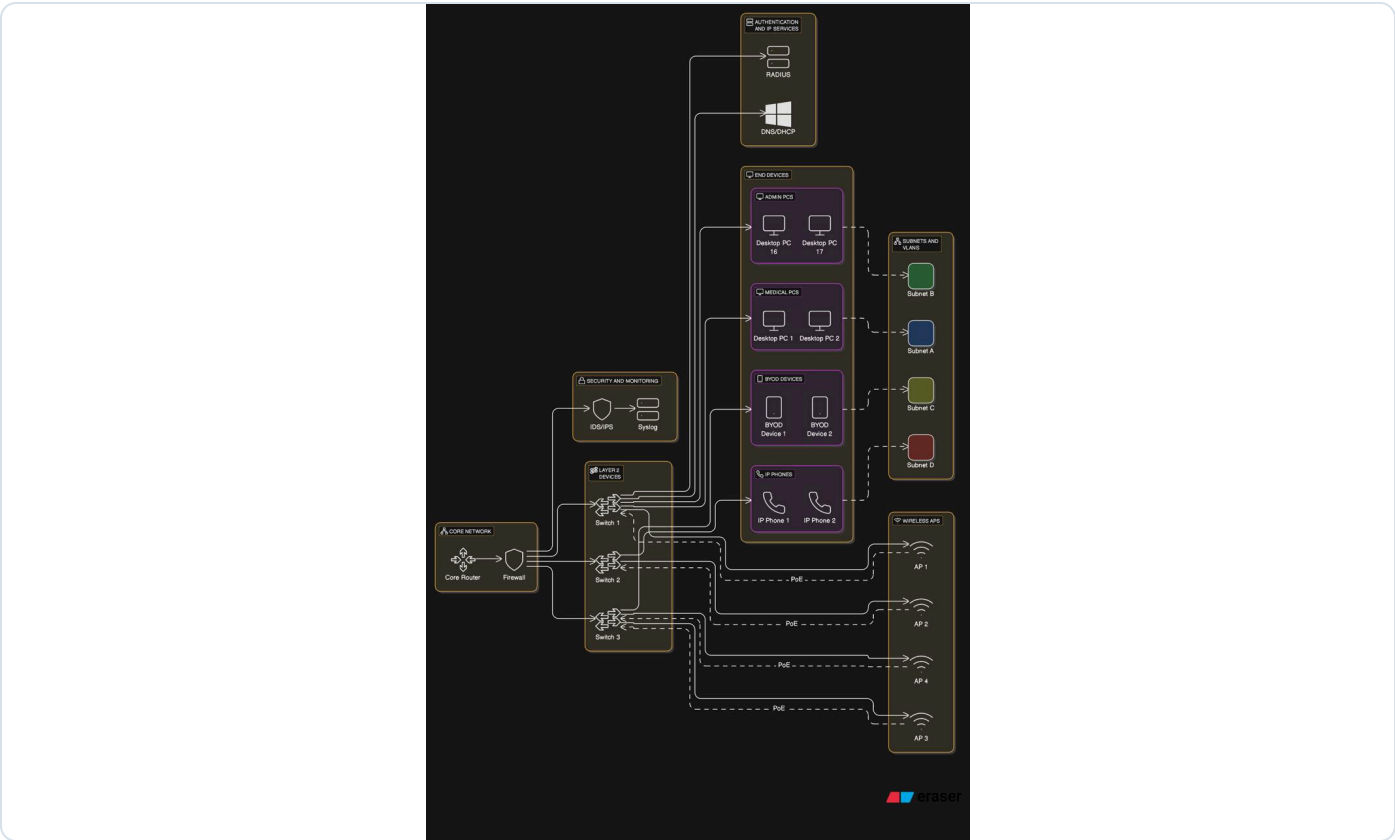


Figure 1. Existing Summit Care logical network material reused as the network context for this security assessment.

Representative Segmentation Model

Segment	Example Subnet	Primary Data / Function	Security Intent
Clinical / EMR	10.7.10.0/24	ePHI, EMR access, charting stations	Restrict lateral movement and limit access to authorized clinical roles.

Segment	Example Subnet	Primary Data / Function	Security Intent
Billing / Finance	10.7.20.0/24	Claims, payment workflows, financial records	Separate payment and billing workflows from clinical and guest traffic.
Administration	10.7.30.0/24	Scheduling, HR, general office workstations	Apply least privilege and monitor document-sharing activity.
Server / Services	10.7.40.0/24	AD, DNS, DHCP, NPS/RADIUS, file services, monitoring	Limit management access; require MFA and admin workstation controls.
Voice / IoT	10.7.50.0/24	VoIP phones, printers, cameras, appliance-like devices	Prevent device networks from reaching ePHI systems except approved paths.
Guest Wireless	10.7.90.0/24	Patient and visitor internet access	Internet-only access with no route to internal clinical resources.

ASSESSMENT METHOD

How The Cyber Threat Analysis Was Conducted

The methodology combines the older cyber threat analysis workflow with the newer Summit Care infrastructure design. It follows a defensible path: identify assets, scan and validate vulnerabilities, test attack paths safely, rate risk, document control gaps, and verify remediation.

- 1 Discovery and asset inventory:** Review diagrams, IP ranges, server roles, endpoint groups, cloud services, access methods, and business processes.
- 2 Threat modeling:** Identify likely adversary behaviors affecting a clinic, including phishing, stolen credentials, exposed remote access, unpatched services, ransomware, and vendor compromise.
- 3 Vulnerability assessment:** Use Nmap, Nessus/OpenVAS-style scanning, configuration review, log review, and manual validation to reduce false positives.
- 4 Controlled penetration testing:** Simulate external, internal, wireless, web application, and privileged-user scenarios without exposing real PHI or disrupting care.
- 5 Risk rating:** Score each finding by likelihood, business impact, data sensitivity, exploitability, and compensating controls.
- 6 Remediation and retest:** Assign owners, set timelines, validate fixes, and document remaining risk accepted by leadership.

Tools Represented

Nmap, Wireshark, Burp Suite, Metasploit in a lab context, Nessus/OpenVAS-style scanners, PowerShell, Python, firewall logs, Windows Event logs, and Microsoft security reporting.

Frameworks Represented

NIST CSF 2.0, HIPAA Security Rule safeguards, OWASP testing concepts, CIS-style hardening practices, and internal risk scoring.

Evidence Types

Asset list, scope memo, scan summaries, sanitized proof screenshots, firewall exports, policy review notes, vendor questionnaires, and retest logs.

ASSET CLASSIFICATION

Critical Systems, Data Types, And Business Impact

Security decisions were prioritized around the systems that store, process, or provide access to ePHI, billing records, user identities, network controls, and recovery data.

Asset / System	Data Type	Business Impact	Baseline Control Requirement
EMR Application / Clinical Workstations	ePHI, patient charts, encounter notes	Critical: direct patient-care disruption and reportable privacy risk.	MFA where supported, role-based access, session timeout, endpoint protection, audit logging.
Billing System / Claims Workflow	Financial records, claim identifiers, limited PHI	High: revenue disruption, fraud exposure, privacy concern.	Web hardening, database least privilege, secure coding, access review, encrypted backups.
Active Directory / Entra Identity	User identities, groups, access policy	Critical: compromise can enable broad lateral movement.	MFA, privileged access workstations, tiered admin groups, lockout policy, audit logging.
Firewall, Switches, Wireless Controllers	Network policy, routes, VLAN enforcement	High: misconfiguration can expose internal systems.	Centralized config backup, named admin accounts, secure management VLAN, change control.
NAS / Backup Repository	System backups, file shares, configuration exports	Critical: ransomware resilience depends on backup integrity.	Immutable or offline copy, restricted admin, encryption, restore testing, separate credentials.

Asset / System	Data Type	Business Impact	Baseline Control Requirement
VoIP, Printers, Cameras, IoT	Operational metadata, possible documents or call records	Medium: pivot risk and privacy-adjacent exposure.	Separate VLAN, default credential removal, firmware updates, egress limits.

THREAT MODEL

Realistic Cyber Threat Scenarios For A Clinic Network

The highest-value risks were not treated as isolated technical bugs. They were analyzed as business scenarios that could affect patient care, PHI confidentiality, billing operations, staff productivity, and regulatory posture.

Scenario A: Credential Theft To EMR Access

High

A targeted phishing email captures a staff password. Without MFA and conditional access, the attacker attempts cloud email access, searches messages for vendor portals, and uses the same identity against VPN or remote desktop services.

Controls: MFA, phishing-resistant training, impossible-travel alerts, passwordless options, log review, and user access reviews.

Scenario B: Ransomware Through Unpatched Endpoint

Critical

An outdated workstation or exposed service allows code execution. The attacker disables local defenses, moves to file shares, encrypts operational data, and pressures the clinic during patient-care hours.

Controls: EDR, patch SLAs, application control, network segmentation, restricted admin rights, immutable backup copies, restore testing.

Scenario C: Billing Portal SQL Injection

High

A vulnerable billing interface accepts unsafe input and allows unauthorized database queries in a test environment. The potential impact includes claim record exposure, financial tampering, and compliance investigation.

Controls: Parameterized queries, secure SDLC, WAF rules, database least privilege, code review, and web application retesting.

Scenario D: Vendor Remote Support Misuse

Medium

A third-party support tool has broad access, inconsistent MFA, or unclear logging. If a vendor account is compromised, the clinic may inherit the attacker path.

Controls: Vendor MFA attestation, named accounts, just-in-time access, session recording, contract security clauses, annual review.

PENETRATION TESTING

Controlled Testing Narrative And Findings

Penetration testing was positioned as validation of risk, not a hunt for dramatic exploits. The test plan measured whether Summit Care's controls could prevent, detect, and contain realistic attack paths.

Testing Phases

Phase	Representative Activity	Security Skill Demonstrated
Reconnaissance	Mapped exposed services, email patterns, DNS records, VPN entry points, and internal subnets after authorization.	Scope control, OSINT discipline, network enumeration, evidence capture.
Vulnerability Validation	Compared scan findings against service versions, patch levels, configuration files, and manual checks.	False-positive reduction, CVSS interpretation, secure configuration review.
Web Application Testing	Reviewed billing and portal workflows for injection risk, authentication weakness, session handling, and error leakage.	OWASP-style testing, Burp proxy workflow, remediation documentation.
Internal Segmentation	Attempted controlled reachability from admin, billing, clinical, IoT, and guest networks to verify firewall intent.	VLAN/ACL testing, least-privilege validation, lateral movement analysis.
Privileged Access Review	Reviewed local admin membership, domain admin exposure, shared credentials, service accounts, and remote management paths.	Identity risk analysis, privilege tiering, Windows security review.

Representative Penetration Test Findings

PT-01: Domain Admin Exposure On Workstations

Critical

Observation: Administrative credentials were used on general-purpose workstations, increasing the chance that endpoint compromise could become domain compromise.

Recommendation: Implement privileged access workstations, remove standing admin access, rotate admin credentials, and enforce tiered administration.

PT-02: Weak SSH / Network Device Management

High

Observation: Network management services were reachable from broader internal segments than required, and older algorithms/default settings were present in the mock review.

Recommendation: Restrict management to a hardened admin VLAN, use named accounts, disable legacy algorithms, and centralize logs.

PT-03: Billing Portal Injection Risk

High

Observation: Testing identified unsafe input handling in a simulated billing workflow, creating a path to unauthorized query behavior.

Recommendation: Use parameterized queries, input validation, code review, least-privilege database accounts, and retest before production release.

PT-04: Guest Network Egress And Discovery Noise

Medium

Observation: Guest Wi-Fi was logically separate from clinical systems, but excessive outbound freedom and device discovery created unnecessary attack surface.

Recommendation: Enforce client isolation, DNS filtering, rate limits, internet-only routing, and captive portal policy acknowledgement.

RISK REGISTER

Prioritized Vulnerability And Risk Assessment Results

Risks were scored using impact and likelihood, with extra weight added for ePHI, identity systems, patient-care availability, and whether the issue could be chained with another weakness.

ID	Risk	Impact	Likelihood	Rating	Owner	Remediation
R-01	Privileged credentials exposed on normal endpoints	5	4	Critical	IT / Security	Admin tiering, PAWs, LAPS/credential rotation, MFA, monitoring.
R-02	Unpatched EMR-supporting web service	5	3	High	Systems	Patch SLA, vulnerability scanning, maintenance windows, compensating WAF rules.
R-03	Billing portal injection weakness	5	3	High	Application / Vendor	Secure coding fix, database least privilege, code review, web retest.
R-04	Remote access policy inconsistent across VPN and cloud tools	4	3	High	Identity	MFA everywhere, conditional access, device compliance, log alerting.
R-05	Third-party remote support lacks	4	3	High	Vendor Management	Vendor questionnaire, MFA proof, BAA

ID	Risk	Impact	Likelihood	Rating	Owner	Remediation
	complete audit evidence					review, access expiration, session logging.
R-06	Backup restore testing not fully documented	5	2	Medium	IT Operations	Quarterly restore test, immutable copy, backup success alerts, runbook update.
R-07	Default or weak SNMP strings on network devices	3	3	Medium	Network	Disable SNMPv1/v2c where possible, enforce SNMPv3, limit source IPs.
R-08	User phishing susceptibility in finance and scheduling groups	4	3	High	Security / HR	Targeted training, simulated campaigns, reporting button, MFA, email filtering.

HIPAA ALIGNMENT

HIPAA Security Rule And Compliance Facts

The HIPAA Security Rule requires regulated entities to protect electronic protected health information with appropriate administrative, physical, and technical safeguards. For this mock clinic, the practical compliance focus is risk analysis, access control, audit controls, integrity, transmission security, workforce security, contingency planning, and vendor oversight.

HIPAA Safeguard Area	Summit Care Control Evidence	Risk Assessment Notes
Administrative Safeguards	Security responsibility assigned, access review process, workforce training, incident response plan, vendor review.	Needs stronger recurring evidence: signed policy acknowledgements, quarterly access reviews, and management acceptance of residual risk.
Physical Safeguards	Server/network closet control, UPS protection, workstation placement, visitor separation, device inventory.	Improve checklist evidence for device disposal, physical access review, and environmental controls.
Technical Safeguards	Unique user IDs, MFA, role-based access, firewall segmentation, encryption, audit logs, endpoint protection.	Most important gaps are privileged access management, log review cadence, and technical proof of backup recovery.
Risk Analysis / Risk Management	Asset inventory, risk register, vulnerability scan summaries, remediation roadmap, retest documentation.	Convert assessment into a living risk register reviewed by leadership each quarter.
Business Associate / Vendor Oversight	BAA tracking, vendor access list, remote support review, third-party questionnaire.	Require proof of vendor MFA, breach notice terms, encryption practices, and access termination process.

Important compliance framing: This report supports a HIPAA-aware security program, but it is not a legal opinion or formal compliance certification. A real covered entity should have

counsel and compliance leadership validate final policies, breach procedures, and regulatory obligations.

THIRD-PARTY RISK

Vendor And External Audit Program

Healthcare clinics rely heavily on vendors: EMR platforms, billing services, VoIP providers, MSPs, backup platforms, cloud storage, email security, internet service providers, and remote support tools. Each vendor can become part of the clinic's attack surface.

Vendor Category	Example Risk	Audit Evidence Requested	Decision
EMR Provider	PHI exposure, uptime dependency, privileged support access.	BAA, SOC 2 or security summary, encryption statement, support access process, breach notification terms.	Monitor
Billing / Claims Vendor	Financial data compromise, web application vulnerability, weak support controls.	Secure SDLC statement, vulnerability management proof, MFA attestation, data retention policy.	Remediate
Managed IT / Remote Support	Broad admin access, shared accounts, incomplete logging.	Named account list, MFA proof, session logs, access expiration process, background check policy summary.	Restrict
Cloud Backup Provider	Backup deletion, account takeover, restore failure.	Immutable backup options, encryption, restore testing logs, admin MFA, data residency statement.	Approve

Third-Party Audit Checklist

- Confirm whether vendor handles ePHI and whether a Business Associate Agreement is required.
- Require named accounts, MFA, and least-privilege support roles for remote access.
- Review breach notification terms, data retention, encryption, backup, and subcontractor language.

- Track corrective actions, due dates, vendor owner, and risk acceptance decisions.
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MONITORING & AUDITING

Routine Security Operations And Evidence Cadence

The older automation and routine auditing content has been converted into an operations model for Summit Care. The point is to make security repeatable: collect logs, check baselines, review access, scan for vulnerabilities, and prove that backups and controls still work.

Daily

Review critical alerts, backup status, endpoint protection status, failed admin logons, and firewall deny spikes.

Weekly

Review vulnerability deltas, firewall changes, high-risk email events, VPN activity, and privileged group changes.

Monthly

Patch servers and network devices, review inactive accounts, validate guest Wi-Fi separation, and export key configs.

Quarterly

Run access recertification, restore-test backups, tabletop incident response, vendor review, and risk register update.

Log Sources To Centralize

Source	Events Of Interest	Security Value
Firewall / VPN	Denied traffic, new remote access sessions, geo-anomalies, rule changes.	Detect perimeter attacks, misuse, and segmentation bypass attempts.

Source	Events Of Interest	Security Value
Active Directory / Entra	Privileged group changes, failed logins, MFA prompts, risky sign-ins.	Detect account compromise and identity drift.
Endpoint Protection	Malware blocks, suspicious scripts, disabled agents, isolation events.	Detect ransomware and endpoint compromise early.
EMR / Billing Applications	Export activity, failed authentication, abnormal access hours, privilege changes.	Support HIPAA audit controls and privacy investigations.

RESPONSE & RECOVERY

Incident Response, Disaster Recovery, And Business Continuity

A security assessment is incomplete unless the organization can respond when a control fails. Summit Care's response model is built around detection, containment, eradication, recovery, communication, and lessons learned.

Ransomware Playbook

Isolate affected VLANs, preserve evidence, disable suspect accounts, validate backup integrity, restore priority services, and notify leadership/compliance.

Lost Device / ePHI Exposure

Confirm encryption and MDM status, remote lock or wipe, assess data involved, document timeline, and trigger privacy review if needed.

Vendor Account Compromise

Disable vendor access, collect session logs, rotate affected credentials, verify systems touched, and require vendor corrective action.

Recovery Priorities

Priority	Service	Target Recovery Approach
1	Identity, DNS, DHCP, network core	Restore foundational access first; validate clean admin credentials and firewall rules.
2	EMR and clinical workstations	Restore patient-care operations using verified backups or vendor continuity procedures.
3	Billing, file shares, printing, voice	Restore operational workflows after clinical systems are stable.

Priority	Service	Target Recovery Approach
4	Guest Wi-Fi, nonessential IoT, secondary services	Return lower-priority services after monitoring confirms clean recovery.

REMEDIATION ROADMAP

30 / 60 / 90 Day Security Improvement Plan

The roadmap translates findings into practical work that a clinic can actually execute. Immediate actions reduce the most dangerous paths; later actions mature monitoring, governance, and audit evidence.

Timeline	Actions	Success Evidence
0-30 Days	Enforce MFA for remote/cloud access, remove shared admin accounts, patch critical systems, isolate guest Wi-Fi, review vendor access, confirm backup success alerts.	MFA report, patch report, updated firewall rules, vendor account list, backup dashboard export.
31-60 Days	Implement privileged access workstation model, database least privilege, SNMPv3, centralized logging, phishing reporting workflow, and secure configuration baselines.	Admin group review, baseline screenshots, SIEM/log dashboard, phishing test metrics.
61-90 Days	Perform retest, restore-test backups, complete HIPAA safeguard evidence binder, run incident response tabletop, complete third-party risk reviews.	Retest letter, restore test record, tabletop after-action report, completed vendor register.
Ongoing	Quarterly vulnerability scans, annual penetration test, monthly access review, continuous monitoring, policy refresh, and leadership risk review.	Recurring audit calendar, risk register updates, board/leadership summary, control exceptions log.

APPENDIX

Evidence Pack, Deliverables, And References

The final security portfolio should include enough evidence for a reviewer to understand the work without exposing live secrets. The items below can be used as a clean index for screenshots, exports, and sanitized technical artifacts.

Recommended Evidence Pack

- Sanitized network diagram and VLAN map.
- Asset inventory and data classification sheet.
- Vulnerability scan summary and false-positive notes.
- Penetration testing scope memo and executive findings.
- Risk register with owners, due dates, and retest status.
- HIPAA safeguard matrix and audit evidence checklist.
- Third-party/vendor risk questionnaire summary.
- Incident response tabletop and backup restore test report.

Key Deliverables Created

- Cyber Threat Analysis and Risk Assessment Master Report.
- Penetration testing narrative and representative findings.
- HIPAA-aware control alignment and compliance evidence plan.
- Third-party audit framework for healthcare vendors.
- 30 / 60 / 90 day remediation roadmap.
- Ongoing monitoring and routine audit cadence.

References Used For Current Compliance Framing

1. HHS, [The Security Rule](#), content last reviewed March 19, 2026.
2. HHS, [Summary of the HIPAA Security Rule](#).
3. HHS, [Security Rule Guidance Material](#).

4. NIST, [The NIST Cybersecurity Framework \(CSF\) 2.0](#), published February 26, 2024.
 5. NIST, [Cybersecurity Framework FAQs](#), describing the six CSF functions: Govern, Identify, Protect, Detect, Respond, Recover.
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